

Locally Plausible, Globally Wasteful: Consensus Consolidation of Shared Experience Silently Degrades Multi-Agent LLM Systems

Anonymous submission

Abstract

Multi-agent LLM systems increasingly share experiential memory: agents write distilled insights into a common pool that is retrieved and injected into every teammate’s context. We study a failure mode of this design that requires no wrong facts, no poisoned text, and no illegal actions. In a controlled trap-maze microscope where every route is legal and shortest-path cost is exactly measurable, we run a preregistered ladder of experiments over faithful instantiations of published components (ExpeL-style insight operations, Generative-Agents-style retrieval). Single-agent faithful experience learning leaves primary efficiency endpoints unchanged, but across a five-arm multi-agent matrix, injected memory degrades efficiency and amplifies looping, and *consensus-consolidated* shared memory is the most reproducibly harmful arm (success-only excess steps +11.8, 95% CI [7.7, 16.0]; loop rate +0.138, CI [0.092, 0.188]; 5/5 seeds) and the only arm whose loop amplification significantly exceeds private memory. The harm is not read-side usage-weighted retrieval (a preregistered null dose-response); it persists without self-healing over a $T=10$ horizon while the append protocol *improves*; and a write-side audit shows consolidation compresses the population’s active strategy supply by $\sim 2.3\times$. Two preregistered rescue interventions bound the mechanism and the obvious fix: re-exposing consolidated-away variants at added dose fails and worsens stagnation, and an equal-dose archive is unreachable under fixed-capacity similarity retrieval. A cross-model probe (Qwen3-32B) structurally replicates the compression but leaves the efficiency effect unresolved and excludes a loop amplification of the primary model’s magnitude. We release the microscope, preregistrations with frozen gates, and per-route artifacts.

1 Introduction

When army ants lose the pheromone trail, each ant follows the locally impeccable rule “follow the ant in front of you,” and the colony organizes itself into a circular mill that marches until exhaustion. No single step is wrong; the composition is fatal.

We present evidence that shared experiential memory in multi-agent LLM systems admits an analogous failure. The

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prevailing design pattern—popularized for single agents by experiential learners such as ExpeL (Zhao et al. 2024) and Reflexion (Shinn et al. 2023), and extended to agent teams by experiential co-learning (Qian et al. 2024a) and cross-task multi-agent experience pools (Zheng et al. 2025)—lets agents distill natural-language insights from their own trajectories, store them in a memory pool, and retrieve them as guidance. When the pool is *shared* and a reviewer LLM *consolidates* it by consensus (merging, editing, up/down-voting insights), every ingredient is individually reasonable and individually published. The composition, we show, is not.

The failure we characterize is *silent*. Agents keep succeeding, never cross walls, never submit false answers; the experience text is locally plausible and derived from genuine trajectories. Yet routes grow longer, agents revisit the same cells, and loop signatures rise—and the effect intensifies exactly under the write protocol that is supposed to improve memory quality. Because no error is ever emitted, success-rate dashboards do not see it.

Studying such a compositional failure on a live benchmark confounds every knob at once. We instead build a *mechanism microscope*: a generated 15×15 trap-maze family with known shortest paths, local observations, and a step cap, in which efficiency loss and looping are exactly measurable per route (Figure 1). Into this microscope we place faithful instantiations of published components (Table 1), so that any degradation is attributable to the composition—sharing and consolidation—rather than to home-made machinery. All confirmatory experiments were preregistered with frozen gates; every deviation and negative result appears in an append-only log.

Our contributions:

- **A preregistered phenomenon.** In a five-arm multi-agent memory matrix over five seeds, consensus-consolidated shared memory degrades success-only route efficiency, amplifies loops, and reduces success relative to a write-only frozen control; storage alone is harmless and multi-agent execution alone is not pathological. Relative to *private* memory, the shared-specific risk is loop amplification, not a broad efficiency gap.
- **Locating the channel.** A preregistered read-side dose-response is null, while a write-side audit shows consolidation compresses the population’s active strategy supply $\sim 2.3\times$. Over $T=10$ the harm persists without self-

healing while the uncompressed append protocol significantly self-improves.

- **Intervention boundaries.** Two preregistered archive-rescue interventions show that the obvious mitigation—keep consolidated-away variants retrievable—fails at added dose and is unreachable at equal dose; supply compression remains a robust correlate whose causal status we report as unresolved.
- **Transfer bounds.** A cross-model probe structurally replicates the write-side compression but not the behavioral coupling, bounding the phenomenon’s transfer instead of assuming it.
- **A reusable microscope and discipline.** We release the environment, runners, paired-bootstrap statistics, sanitizer, memory audits, and preregistrations with freeze hashes.

2 Related Work

Single-agent experiential learning. ExpeL (Zhao et al. 2024) distills insights from success/failure trajectories with ADD/EDIT/UPVOTE/DOWNVOTE operations and retrieves them as guidance; Reflexion (Shinn et al. 2023) converts failures into verbal self-corrections; ReAct (Yao et al. 2023) interleaves reasoning and acting; Generative Agents (Park et al. 2023) score memories by relevance, importance, and recency; MemoryBank (Zhong et al. 2024) adds long-term memory management. These works established that experience helps a *single* agent. Our write operations, retrieval scoring, and distillation prompts are faithful reimplementations of this line (Table 1); we stress-test their composition, not their original claims.

Multi-agent experience sharing. Experiential co-learning (Qian et al. 2024a) shares shortcut experiences between instructor and assistant agents in ChatDev (Qian et al. 2024b); MAEL (Zheng et al. 2025) equips each agent in a collaboration graph with an experience pool queried by similarity and reward; multi-agent debate (Du et al. 2024) and self-consistency (Wang et al. 2023) aggregate opinions rather than memories; MetaGPT (Hong et al. 2024) shares a message pool. All published sharing designs accumulate experience additively. To our knowledge, no prior work evaluates the natural next step—*LLM-driven consensus consolidation of a shared experience pool*—with controls; that protocol is exactly our subject.

Memory risk and governance. A concurrent line studies shared-memory governance: failure taxonomies for multi-tenant memory (Anonymous 2026a), stress-tests of memory systems (Anonymous 2026c), drift and poisoning risks (Anonymous 2026b), and longitudinal safety risks of memory-equipped agents (Anonymous 2026d). These works propose frameworks and taxonomies over deployed or synthetic traces. We complement them with a preregistered, controlled, intervention-bearing study of one specific mechanism—consolidation-induced degradation—with exact route-level ground truth, and with negative results that bound the cheapest mitigation.

Table 1: Every component is a faithful instantiation of a published method; only the composition (shared pool + consensus consolidation) is new, and it is the subject of study.

Component		Source
Insight ops	ADD/EDIT/UP/DOWN, importance-0 deletion	ExpeL
Retrieval	scoring (relevance+importance+recency, λ)	Generative Agents
Insight distillation from trajectories		ExpeL
Stepwise solving with injected experience		ReAct-style
Shared pool across agents		Co-learning, MAEL
Consensus consolidation of the shared pool	composition	under test

3 The Maze Microscope

Environment. We generate a 15×15 “trap” maze family with enforced shortest-path length ≥ 30 and disjoint train/heldout splits per experiment seed. Solvers receive local observations only (no maze graph, no shortest path), act under a 120-step cap, and are scored by exact excess steps over the known optimum. Efficiency loss and loop signatures are therefore measurable per route—the property that makes silent degradation visible at all.

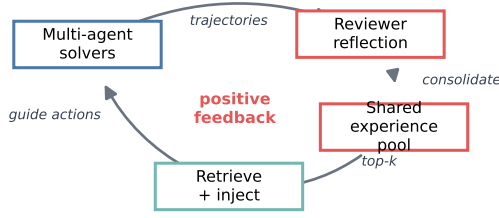
System under test. Four solver agents attempt the *same* tasks in independent environment copies. They never communicate and never vote; the *only* coupling channel is the experience memory, so any cross-agent effect is attributable to it. Each round t comprises (i) evaluation on 12 held-out mazes per agent (48 routes), (ii) a training batch of 4 mazes per agent (16 routes), and (iii) a reviewer LLM that sees only logs and quality signals—never hidden routes—and writes to the memory pool under the arm’s protocol. The final round evaluates without writing. Retrieval injects the top- $k=6$ insights (Generative-Agents scoring; $\lambda=0$ outside the dose-response experiment); the pool is capped at 80.

Arms. The five-arm matrix decomposes “multi-agent shared consolidated memory” factor by factor: *no-memory* (is multi-agent execution itself pathological?); *frozen* (reviewer writes, solvers never read: is *storage* harmful?); *private* (per-agent pools: is *sharing* necessary?); *shared-append* (shared but never merged: is sharing without consolidation harmful?); and *shared-consolidated* (the pattern under indictment).

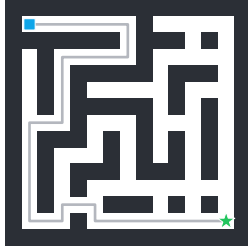
Endpoints and statistics. Primary preregistered endpoints: success-only excess steps, loop rate, and the multi-agent ant-mill rate (loop + route overlap). Auxiliary: success, stagnation, maximum revisits, cost ratio, route diversity. Mechanism audit, logged per run: pool-size trajectory, *distinct injected experiences*, retrieval entropy and top-1 share, sanitizer rejections, LLM errors. All verdicts use same-maze same-agent paired bootstrap (10k resamples, 95% CI) with per-seed sign tables. A sanitizer rejects maze IDs, coordinates, and fixed action sequences from memory text; cache keys are salted by (run, seed, round, task, agent, step). The

Figure 1 | Consensus consolidation turns correct shared experience into a silent ant-mill loop

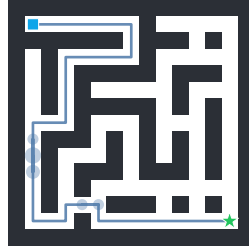
a Shared-experience feedback loop



Frozen write-only
steps 39 · cost 1.03
max revisit 1



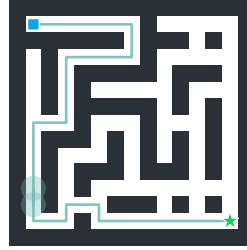
Private memory
steps 47 · cost 1.24
max revisit 4



b Silent degradation on one held-out maze

Same maze, same agent, final round (seed 4). Every route is legal — no wall crossing, no false submit — and every arm still reaches the goal. As the write protocol consolidates experience, the agent revisits more cells and finally loops, while shortest path stays 38 steps.

Shared append
steps 55 · cost 1.45
max revisit 9



Shared consolidated
steps 73 · cost 1.92
max revisit 11 · LOOP

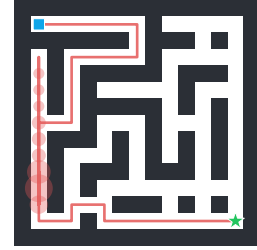


Figure 1: Mechanism microscope and a silent degradation case. (a) The shared-experience feedback loop: solver trajectories are reflected into a shared pool, consolidated by LLM-issued operations, retrieved top- k , and injected back as guidance—a positive-feedback cycle. (b) Same held-out maze and agent, final round; shortest path 38. Revisit heat grows as the write protocol consolidates, culminating in a silent loop under shared-consolidated memory, while every route stays legal and successful.

primary model is DeepSeek-V3 (DeepSeek-AI 2024) via a fixed endpoint; a preregistered drift probe (cache disabled, replaying a frozen-arm round; all probe metrics inside the original bootstrap CIs) authorized reuse of cached controls.

4 Preregistered Results: the Phenomenon

4.1 E1: Single-Agent Attribution Control

Before introducing sharing, we ask whether faithful experience learning changes single-agent behavior at all. Relative to a no-memory baseline (paired, final round, three seeds), neither the ExpeL-operations writer nor the append writer moves any primary endpoint: success-only excess steps CIs contain zero (append -4.74 , CI $[-14.1, +4.2]$; ops $+5.11$, CI $[-6.4, +16.3]$), as do loop rates. By the frozen rule this is a **fail**, recorded as such. However, the mandated adherence analysis shows experience is *not* ignored: the ops writer shifts action selection away from the controller suggestion after memory activates ($+0.041$ deviation, same sign in 3/3 seeds), and the append writer significantly reduces maximum revisits. The single-agent null is therefore an *attribution control*: any multi-agent degradation cannot be attributed to “using memory” per se, only to the sharing channel.

4.2 E2: Multi-Agent Memory Matrix

We compare the five arms at four agents, $T=6$, five seeds, against the write-only *frozen* control. Table 2 and Figure 2 report paired differences at the final round.

Table 2: E2 final-round paired differences versus frozen write-only memory (five seeds, $n_{\text{pairs}} = 240$; success-only excess steps drops unpaired failures). * = 95% CI excludes zero.

Arm	Succ.	Excess	Loop	Revisit
MAS no memory	-0.004	$+1.12$	$+0.042^*$	$+0.10$
Private	-0.175^*	$+7.32^*$	$+0.079^*$	$+1.20^*$
Shared append	-0.121^*	$+2.75$	$+0.025$	$+0.23$
Shared consol.	-0.171^*	$+11.84^*$	$+0.138^*$	$+1.68^*$

Three readings follow. First, multi-agent execution is not itself pathological: MAS-no-memory matches frozen on efficiency and success, with only a small loop increase ($+0.042$, CI $[0.008, 0.075]$); frozen itself matches no-memory, so *storage is harmless and injection is the harm channel*. Second, **consensus-consolidated shared memory meets every preregistered degradation criterion**: success-only excess steps $+11.84$ (CI $[7.66, 16.02]$), same-sign in 5/5 seeds and *larger* in the later seeds; loop rate $+0.138$ (CI $[0.092, 0.188]$); success -0.171 . Third, the broad risk is active injection rather than sharing alone—private memory also degrades—so we test the sharing-specific claim directly.

Shared versus private boundary. Figure 3 contrasts the shared arms against a *private* baseline. The efficiency separation is not significant (consolidated vs. private success-only excess steps $+0.87$, CI $[-4.29, +6.15]$), but loop ampli-

E2: active consolidated memory degrades efficiency and increases loops

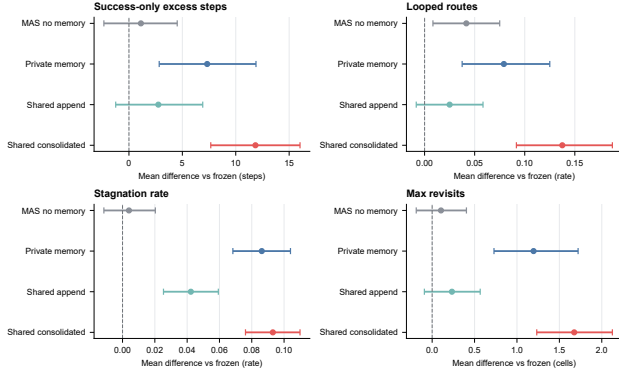


Figure 2: **E2 main result: paired differences versus frozen write-only memory** (five seeds, final round). Points are mean differences; bars are 95% bootstrap CIs; the dashed line is zero. Shared-consolidated (red) degrades on all four end-points with the largest effects; MAS-no-memory (grey) sits on the zero line except for a small loop increase.

E2 boundary: vs private memory, only looping separates (filled = 95% CI excludes 0)

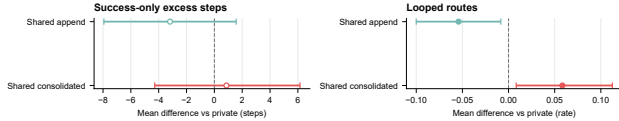


Figure 3: **E2 boundary: shared arms versus a private baseline**. Filled markers mark CIs that exclude zero. Only looping separates shared-consolidated from private (+0.058, CI [0.008, 0.113]); the efficiency contrast does not, and shared-append loops less than private.

cation is: consolidated loops significantly more than private (+0.058, CI [0.008, 0.113]), while append loops *less*. The honest claim is therefore that consensus consolidation specifically amplifies looping relative to private memory; we do not claim a broad shared-over-private efficiency gap.

4.3 E3: Read-Side Dose-Response Is Null

If the harm were read-side—agents increasingly retrieving a few high-vote memories—then raising the importance weight λ in Generative-Agents-style scoring should deepen degradation. It does not. Sweeping $\lambda \in \{0, 0.5, 1\}$ on the append arm (three seeds, $T=6$), the paired difference of $\lambda=1$ versus $\lambda=0$ on success-only excess steps is +1.69 (CI [-4.26, +7.61]), loop rate +0.049 (CI [0.000, 0.104]), and final-round retrieval entropy is non-monotone in λ (0.833/0.826/0.845). The read-side usage-weighted-feedback hypothesis is refuted; we report it as a clean null.

4.4 Mechanism Audit: Write-Side Strategy-Supply Compression

An offline audit of the same E2 runs locates the concentration on the *write* side (Figure 4). Under identical settings, the

consolidated pool converges to 14.2 items with only 11.0 distinct experiences ever injected across all rounds, whereas the append pool grows to the cap (80) and injects 25.4 distinct experiences, and private injects 37.6. Consensus consolidation compresses the population’s active strategy supply by roughly $2.3\times$ relative to append. The top-1 retrieval share rises in step, but this is a consequence of the smaller pool rather than an independent channel; the load-bearing evidence is the direct pool-size and distinct-injection counts. Given the intervention results below, we report compression as a robust, cross-checked *correlate* and candidate mechanism, not as established cause.

4.5 E4: Long-Horizon Stress Test (Exploratory)

To ask whether the write-side harm saturates, deepens, or self-heals over a longer feedback horizon, we extend the four-agent shared arms to $T=10$ (three seeds, heldout 12, frozen and shared-append as controls; not gate-bound, per the preregistration). Relative to frozen, shared-consolidated’s degradation is present at every checkpoint tested ($t=3, 6, 9$) and does not narrow: success drops $-0.174, -0.146, -0.257$ (all CIs exclude zero) and loop rate rises $+0.083, +0.083, +0.118$ (all CIs exclude zero), with the final round the worst of the three. A stricter within-condition test—each condition’s last three rounds against its first three (excluding warm-up)—finds no significant self-trend for shared-consolidated on any of six endpoints: the gap opens early and persists at a noisy plateau rather than provably deepening. The append arm, in contrast, shows a significant within-condition *improvement* over the same horizon (cost ratio -0.079 , excess steps -3.50 , stagnation -0.026 , max revisits -0.340 , all CIs exclude zero) and by $t=9$ is significantly cheaper than frozen on three endpoints. This contrast is also the paper’s validity anchor: the same read/write infrastructure that degrades under consolidation *helps* under append, so the harm cannot be blamed on “memory” or on our plumbing. Per-seed decomposition exposes heterogeneity the pooled estimate hides: seed 1 shows the only significant late-vs-early decline in success (-0.097); seed 2 has the most severe round-wise trajectory (ending at success 0.17, loop rate 0.46) though its late-vs-early CI includes zero; seed 0 shows no decline. The pooled “no significant further drift” verdict is a mean over a mixed population, not evidence that every instance is equally stable.

4.6 The Failure Is Silent

Figure 1(b) shows a case study mined from the runs: the same held-out maze and agent at the final round, whose shortest path is 38 steps. As the write protocol consolidates, the route degrades monotonically—frozen 39 steps (one revisit), private 47 (four), shared-append 55 (nine), shared-consolidated 73 with cost 1.92, eleven revisits to the same cell, and a detected loop—yet every route is legal and every arm still reaches the goal. This is the ant-mill signature: not a false belief, but a locally plausible correction signal, reinforced through consensus writing, that composes into a globally wasteful cycle.

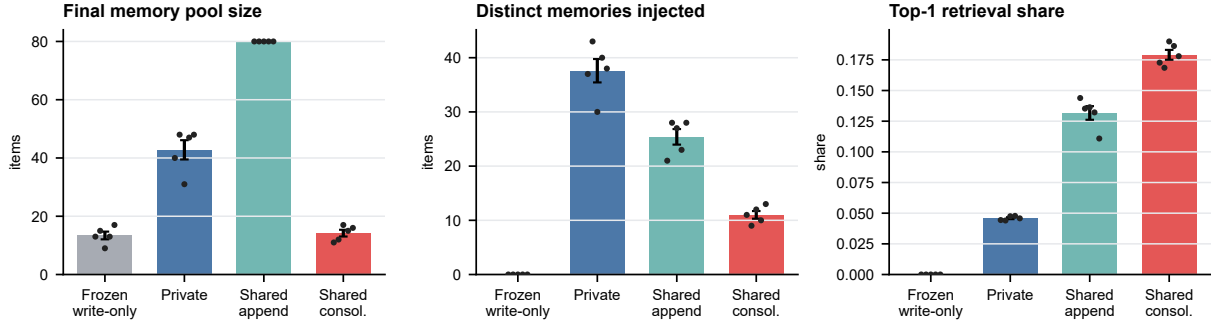


Figure 4: **Mechanism audit: consolidation compresses the strategy supply.** Final pool size, distinct experiences ever injected, and top-1 retrieval share across five seeds (dots). Consensus consolidation keeps a small pool and injects the fewest distinct strategies, concentrating retrieval; the direct counts (left, centre) carry the claim.

Table 3: Preregistered rescue interventions on the consolidated arm (three seeds each). Neither supports promoting supply compression from correlate to cause; both bound the “keep an archive” mitigation.

Arm	Manipulation	Behavioral outcome
Additive archive (top-6 + archive top-6; dose $2\times$)	supply (21/21/27 vs. 12/11/14)	no endpoint improves; stagnation worsens $+0.045^*$
Equal-dose joint ranking (total top-6)	supply 1.42/1.55/1.07 $\times$: check fails	not evaluable by frozen rule

5 Mechanism Interventions and Boundaries

A second preregistration (Phase Gamma) froze three short, gated extensions: two archive-rescue interventions on the primary model and a cross-model probe (Table 3).

5.1 Rescue I: Additive Archive Access Fails and Backfires

The cheapest conceivable mitigation makes consolidation non-destructive: retain DOWN-voted insights, pre-edit versions, and similarity-merge-discarded variants in a read-only *archive* that solvers can still retrieve. Our first preregistered rescue arm implemented archive access additively (active top-6 *plus* archive top-6), which doubles the injected dose to 12 items per prompt—a deviation we record and exploit. The manipulation itself succeeded: distinct injected experiences roughly doubled, with archive items winning about half the retrieval slots. Behavior did not improve on a single endpoint: success non-inferiority failed (-0.056 , CI $[-0.153, +0.042]$ against a -0.05 bound); success-only excess steps were null (-0.45 , CI $[-6.86, +5.96]$); loops trended worse ($+0.035$); stagnation significantly worsened ($+0.045$, CI $[+0.023, +0.067]$). Read as a designed-fix evaluation: naively re-exposing the material consolidation discarded, at added dose, does not rescue behavior and aggra-

vates stagnation—consistent with E2’s finding that injection itself is the harm channel.

5.2 Rescue II: the Equal-Dose Archive Is Unreachable

The confounded dose motivated a second, clean arm: one *joint* ranking over active pool and archive with the original total top-6, leaving the per-prompt dose identical to E2. The preregistered manipulation check required the cumulative distinct injected supply to expand by $\geq 1.5\times$ in at least 2/3 seeds. It did not (ratios 1.42/1.55/1.07), and not because the archive was ignored—archive items appeared in 896/912 retrievals where candidates existed. Under a fixed retrieval capacity, consolidated insights dominate the similarity ranking, so the archive substitutes items without expanding *diversity*. Per the frozen amendment, the behavioral gate was not evaluated, and we explicitly do *not* claim that equal-dose archive rescue is ineffective. The constructive reading: the popular “never really delete; keep an archive” design does not restore strategy diversity under fixed-capacity similarity retrieval—a reachability bottleneck any governance proposal must confront. Together the two rescue arms leave strategy-supply compression a strong correlate whose causal role is unresolved; we report this plainly rather than promoting the mechanism.

5.3 Cross-Model Probe: Compression Replicates, Coupling Does Not

We repeated the frozen/append/consolidated triangle on Qwen3-32B (Qwen Team 2025) (three seeds, $T=4$, justified by E4’s finding that the primary-model gap is established by $t=3$; thinking mode disabled after a passing calibration smoke: 100% action parse rate, baseline success 0.479). The preregistered gate reports *not-detected-or-underpowered*, but the three endpoints earn three different honest readings (Figure 5):

- *Efficiency: unresolved, direction-consistent.* Success-only excess steps $+5.83$ with all three seeds positive ($+12.4/+$

0.9/+1.8), but CI $[-3.75, +15.20]$ ($n=40$ success pairs) contains both zero *and* the DeepSeek effect (+11.84): the probe cannot adjudicate between no effect and a full-size effect. Power was consumed by Qwen’s lower baseline success, which shrinks the success-only pair count.

- *Loops: a bounded difference.* Loop rate -0.076 , CI $[-0.153, 0.000]$: the CI *excludes* the primary model’s $+0.138$. A loop amplification of that magnitude is absent on Qwen at these settings. We report this as an exploratory magnitude comparison, not a preregistered boundary claim.
- *Success: no drop.* $+0.042$, CI $[-0.049, +0.132]$, excluding the primary model’s -0.171 .

Crucially, the *mechanism precondition* does transfer: at the matched horizon ($t=3$), consolidation compresses the distinct injected supply on Qwen by $1.55\times$ (13.3 vs. 20.7) versus $1.68\times$ on DeepSeek—nearly identical compression, different behavioral coupling. Structural compression without detected harm on one model further cautions against treating compression alone as sufficient cause, consistent with the rescue results above.

6 Design Implications

- **Do not consensus-consolidate shared experience by default.** The same infrastructure that degrades under consolidation improves under append-and-retrieve (E4’s significant within-condition gains). If dedup or cap pressure demands consolidation, treat it as a risk surface, not hygiene.
- **Monitor the write side, not just success.** Distinct injected experiences, pool-size trajectory, and retrieval concentration are cheap, logged-at-source early-warning signals; success-rate dashboards are blind to silent efficiency decay by construction.
- **Loop and stagnation signatures belong in agent telemetry.** Repeated state-action signatures and stagnation rate separated consolidated from private memory when aggregate efficiency could not.
- **“Keep an archive” is not a free fix.** At added dose it backfires; at equal dose fixed-capacity similarity retrieval never reaches the archive. Governance proposals that retain pruned memories must also change the *reachability* of what they retain.

7 Limitations

Our confirmatory evidence is single-environment and single-model-family: a controlled maze microscope on DeepSeek-V3, chosen because its difficulty band makes silent degradation exactly measurable and because inexpensive open models are precisely what shared-memory multi-agent deployments run at scale; the precedent for mid-tier backbones in this literature is explicit (Zhao et al. 2024). The cross-model probe bounds but does not establish transfer: efficiency is unresolved (underpowered) and the loop signature does not carry at the primary model’s magnitude. The compression mechanism remains a candidate—our own interventions constrain it honestly. Task-family generalization

(MiniWoB (Shi et al. 2017; Liu et al. 2018) via Browser-Gym (de Chezelles et al. 2024)) is preregistered and in progress after an action-interface repair; tool-workflow domains such as τ^2 -bench (Barres et al. 2025) remain future work. Effects may differ with stronger reviewers (a consolidator-capability ablation is natural follow-up), other consolidation prompts, or retrieval schemes beyond similarity ranking.

8 Conclusion

Shared experiential memory is being adopted faster than it is being audited. We showed, under preregistered gates in a measurable microscope, that the most natural “quality-improving” write protocol—LLM consensus consolidation of a shared pool—is the most reproducibly harmful one: it silently taxes route efficiency, amplifies loops beyond private memory, locks in early rather than self-healing, and compresses the population’s strategy supply, while the humbler append protocol keeps improving with the same plumbing. The cheapest fixes do not work, and the phenomenon’s transfer across models is bounded rather than assumed. The ant-mill is a useful warning: nothing needs to be wrong for everything to go in circles.

Reproducibility. All preregistrations (with SHA-256 freeze records), decision logs, per-route artifacts, memory audits, statistics code, and the maze/browser runners are released; every number in this paper maps to a frozen gate report or an explicitly labeled exploratory analysis.

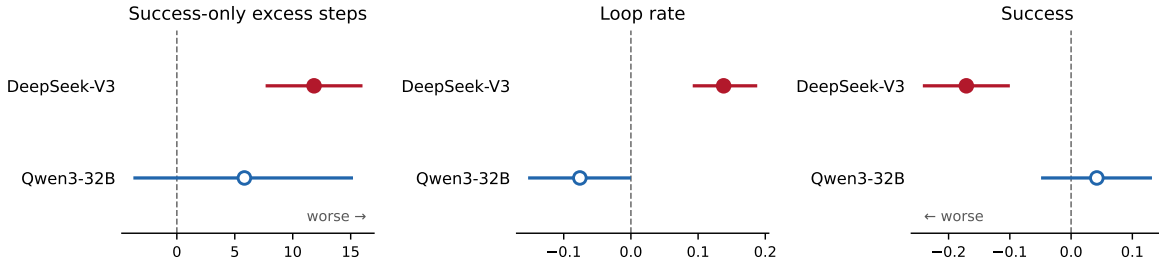


Figure 5: **Cross-model probe: consolidated minus frozen, paired bootstrap 95% CIs.** Filled markers: CI excludes zero. Efficiency (left) is direction-consistent on Qwen but its CI covers both zero and the DeepSeek effect (underpowered); the Qwen loop CI (centre) excludes a DeepSeek-magnitude amplification; success (right) shows no drop on Qwen.

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